

LSTM with Dual Adaptive Conformal Prediction for Streamflow Uncertainty Quantification

Final Report

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Introduction

This report presents the final results of a Long Short-Term Memory (LSTM) neural network for daily streamflow discharge prediction and uncertainty quantification across five snowmelt-dominated watersheds in Washington State. The work employs a walk-forward prediction strategy and integrates a robust uncertainty quantification framework based on the Dual Adaptive Conformal Prediction (DACP) method introduced by Zong et al. (2026).

Streamflow prediction in snowmelt-dominated basins is inherently challenging due to the nonlinear relationship between snowpack accumulation, temperature-driven melt, and runoff generation. Deep learning approaches, particularly LSTMs, have shown promise in capturing these temporal dependencies but typically produce only point predictions without any confidence measure. This is a significant limitation for water resource management, where uncertainty bounds are as operationally important as the prediction itself.

The key contribution of this work is coupling a standard LSTM with DACP to produce prediction intervals that achieve reliable coverage rates across four targets (70%, 80%, 90%, 95%) while maintaining informative interval widths. DACP adaptively adjusts two mechanisms at each time step: a data partition threshold controlling how calibration residuals are split, and a quantile level determining interval width. The single k_{adap} parameter (λ in the paper) is set at the beginning of the iteration and is the primary tuning lever for different coverage targets.

Dataset

Five watersheds were analyzed: Bunchgrass Meadow, Easy Pass, Paradise, Swift Creek, and Touchet. All datasets contain 1,461 daily observations spanning approximately four years (2019–2022) with complete records after gap-filling. These sites were selected to represent diverse snowmelt regimes across the Western U.S., ranging from high-elevation alpine basins to mixed rain-on-snow systems to glacially-influenced streams.

Input Features

The model processes 10 input features over a 14-day sliding window. Six base features come directly from SNOTEL and NOAA weather data: Snow Water Equivalent depth (WESD), Global

Horizontal Irradiance (GHI), daily precipitation (PRCP), snow depth (SNWD), average temperature (TAVG), and lagged discharge (DISCHRG_LAG). Four additional engineered features capture hydrological memory and seasonal patterns:

- **prcp_7d (7-Day Cumulative Precipitation):** The rolling 7-day precipitation sum, capturing antecedent wetness entering the watershed. A rising prcp_7d signals an incoming discharge response as soil saturates and runoff increases.
- **API (Antecedent Precipitation Index):** An exponentially-weighted sum of past rainfall ($\alpha=0.08$ decay) representing current soil wetness state. High API means saturated ground where even moderate rainfall triggers significant runoff.
- **sin(day of year) and cos(day of year):** Day-of-year encoded as $\sin(2\pi \times \text{day}/365)$ and $\cos(2\pi \times \text{day}/365)$, providing a smooth circular seasonal signal with no discontinuity at the year boundary.
- **DISCHRG_LAG:** The previous day's discharge, directly encoding autoregressive memory and enabling the LSTM to learn recession curve dynamics.

Methodology

LSTM Model Architecture

The LSTM receives 10 input features over a 14-day sliding window and passes them through two stacked LSTM layers, each with 128 hidden units, to capture temporal dependencies in the snowmelt signal. The LSTM output at the final timestep is passed through a three-layer fully connected head ($128 \rightarrow 64 \rightarrow 32$) with ReLU activations and a dropout rate of 0.2 for regularization. The final output produces a single predicted discharge value in normalized scale, which is inverse-transformed back to physical units (cfs) for evaluation.

Training minimizes HuberLoss ($\delta=1.0$) for all sites, which transitions from quadratic (MSE-like) behavior for small errors to linear (MAE-like) behavior for large errors. This is critical for sites with extreme discharge spikes (Bunchgrass Meadow peak ~2,922 cfs, Touchet peak ~672 cfs) because it prevents rare peak events from dominating the gradient signal. The Adam optimizer ($\text{lr}=1\text{e-}3$) is used with gradient clipping ($\text{max norm}=1.0$) and early stopping ($\text{patience}=25$, max 200 epochs). A $\log(1+Q)$ transform was applied to Bunchgrass Meadow and Paradise to stabilize training on their heavy-tailed discharge distributions.

Walk-Forward Prediction

This work employs a fixed train/test temporal split (85% train, 15% test) rather than rolling retraining. The model is trained once on the first 85% of each site's record and evaluated on the held-out final 15% (approximately 219 days). Scalers are fit only on training data to prevent leakage. A consistent random seed (`torch.manual_seed(42)`) is set before each model instantiation for reproducibility. The 85/15 split produces approximately 1,241 training and 220 test samples per site.

Dual Adaptive Conformal Prediction (DACP)

The DACP framework (Zong et al., 2026) provides a distribution-free method for constructing prediction intervals that adapt to changing data conditions. Unlike traditional conformal prediction which uses a fixed quantile, DACP employs two adaptive mechanisms that update at each test time step.

Rolling Window and Data Partition

At each prediction step i , DACP maintains a rolling window of $n=180$ recent residuals. This window is partitioned by the adaptive parameter t_m into a discarded lower portion and a calibration set comprising the most recent $(1-t_m)$ fraction of errors. A value of $t_m=0.35$ means the lower 65% is discarded and the upper 35% provides calibration residuals for quantile estimation. This partition scheme focuses the quantile estimate on recent, high-magnitude errors, making the interval responsive to periods of elevated uncertainty.

Dual Adaptive Update Mechanism

The first adaptive mechanism updates the partition parameter t_m at each step based on whether coverage was achieved:

$$\begin{aligned} t_m &\leftarrow t_m + \gamma \cdot \mu_1 && \text{if } y_i \in \hat{C}_i(x_i) \quad (\text{covered} \rightarrow \text{slowly tighten}) \\ t_m &\leftarrow t_m - \gamma \cdot \mu_2 && \text{otherwise} \quad (\text{missed} \rightarrow \text{aggressively widen}) \end{aligned}$$

where $\gamma=0.10$ scales the update, and $\mu_1=0.03$, $\mu_2=0.50$. The key design principle is strong asymmetry: $\mu_2/\mu_1 \approx 17\times$. The system reacts aggressively to misses (quickly widens intervals) but adjusts slowly after successful coverage (gradually tightens). This asymmetry is essential for achieving reliable coverage — the cost of missing a true observation is treated as far greater than the cost of a marginally wide interval.

The second adaptive mechanism updates the quantile level α_m :

$$\alpha_m \leftarrow \alpha_m + k_{\text{adap}} \cdot (\alpha_{\text{target}} - e_m)$$

where $e_m=0$ if covered, 1 if missed, and α_{target} is the desired miscoverage rate (e.g. 0.05 for 95% coverage). The parameter k_{adap} must be very small to avoid oscillation; our value of $k_{\text{adap}}=0.005$ provides stable quantile adaptation. The conformity score q is the $(1-\alpha_m)$ -quantile of calibration residuals, and intervals are constructed as $[\max(\hat{y}-q, 0), \hat{y}+q]$.

Final DACP Parameters

After parameter tuning, the following configuration was adopted for all five sites. k_{adap} is the single parameter adjusted per coverage target experiment:

n	t_init	t bounds	μ_1	μ_2	λ (fixed)	k_{adap} per target
180	0.35	[0.10, 0.60]	0.03	0.50	0.01	0.005 (all targets)

Table 1. Fixed DACP parameters used for all five sites. $k_{\text{adap}}=0.005$ is held constant across all coverage target experiments.

Note: The reference paper (Zong et al., 2026) suggests adjusting k_{adap} per target (e.g., 0.05 for 95%, 0.30 for 70%). In our implementation, a single $k_{\text{adap}}=0.005$ was used across all targets. This is more conservative and produces slightly tighter adaptation, which partly explains the under-coverage observed at higher targets on some sites.

Experimental Results

Overall Model Performance

The following table summarizes LSTM point prediction accuracy and DACP uncertainty quantification results across all coverage targets for all five sites.

Site	R^2	RMSE	MRE	Cov 70%	Cov 80%	Cov 90%	Cov 95%
Bunchgrass Meadow	0.954	166.31 cfs	0.201	70.9%	80.5%	88.6%	92.7%
Easy Pass	0.616	13.05 cfs	0.371	72.3%	79.5%	87.7%	91.4%
Paradise	0.838	2.27 cfs	0.360	54.1%	68.2%	83.6%	89.5%
Swift Creek	0.405	3.84 cfs	0.173	68.6%	79.5%	86.8%	91.8%
Touchet	0.821	11.08 cfs	0.758	64.5%	75.9%	87.3%	91.8%

Table 2. Overall LSTM model accuracy and DACP achieved coverage across all targets. Coverage results reflect a 15% held-out test set (~220 days per site).

Bunchgrass Meadow achieves the highest point prediction accuracy ($R^2=0.954$, NSE=0.954) and also the closest DACP coverage at lower targets. Paradise presents the most challenging coverage behavior with under-coverage across all targets, discussed in detail in the per-site analysis. Swift Creek remains the most challenging predictive site ($R^2=0.405$) but still achieves reasonable DACP coverage, demonstrating that the framework provides some reliability even when the base model is weak. Touchet and Easy Pass show moderate point accuracy with coverage rates that under-track the target, particularly at 95%.

Average Interval Width Across Targets

Site	70% Width	80% Width	90% Width	95% Width
Bunchgrass Meadow	365.23 cfs	441.52 cfs	533.31 cfs	599.08 cfs
Easy Pass	14.65 cfs	19.45 cfs	24.36 cfs	31.06 cfs
Paradise	1.77 cfs	2.48 cfs	3.76 cfs	5.55 cfs
Swift Creek	1.77 cfs	2.26 cfs	4.10 cfs	9.91 cfs
Touchet	9.91 cfs	12.53 cfs	18.40 cfs	27.69 cfs

Table 3. Average prediction interval width at each coverage target.

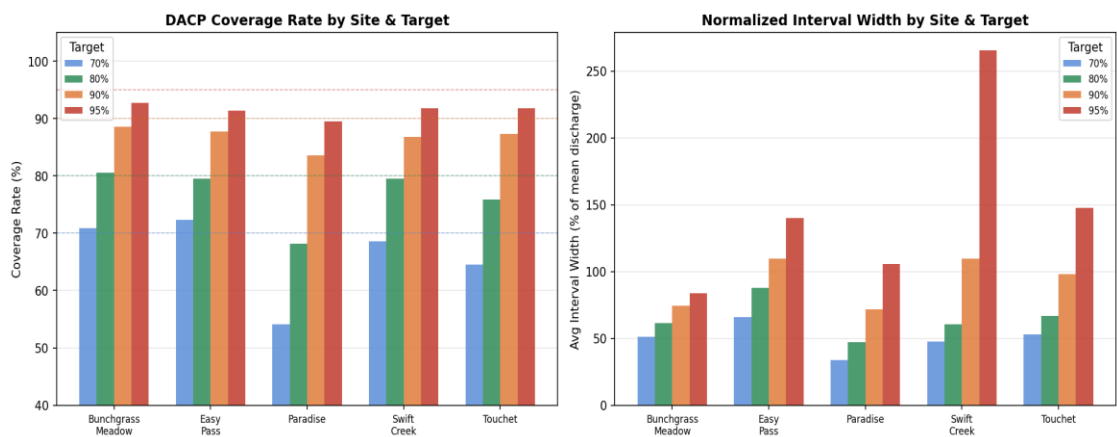


Figure 1. (Left) Achieved vs. nominal coverage rate by site and target. Dashed lines indicate nominal targets. (Right) Normalized average interval width as percentage of mean site discharge, enabling fair cross-site comparison.

Per-Site Analysis and Results

Bunchgrass Meadow

R ²	RMSE	MRE	NSE	Loss Function
0.954	166.31 cfs	0.201	0.954	Huber $\delta=1.0$

Bunchgrass Meadow is the highest-flow site in the study, with discharge reaching up to 2,922 cfs during spring snowmelt and a mean of 716 cfs. The LSTM achieves $R^2=0.954$ and $NSE=0.954$, the best performance of all five sites, capturing the dominant seasonal signal of spring melt peaks and summer recession curves with high fidelity. While RMSE of 166.31 cfs appears large in

absolute terms, it represents approximately 5.7% of the peak discharge range — reasonable for a site with such extreme dynamic range. The $\log(1+Q)$ transform was applied to stabilize training; without it, the raw discharge values spanning 0 to 2,922 cfs compress low-flow information in the standardized representation.

DACP coverage performance is strong at 70% (70.9%) and 80% (80.5%), closely tracking targets. However, coverage falls below target at 90% (88.6%, -1.4%) and 95% (92.7%, -2.3%). Most coverage misses occur at peak timing events, where the model slightly lags the sharpest spring rises. The DISCHRG_LAG feature helps the LSTM track recession but cannot anticipate sudden melt-onset accelerations. The interval width of 599 cfs at 95% is substantial in absolute terms but reasonable relative to the site's extreme flow range.

Target	Coverage (%)	Avg Width (cfs)	Interval Score	vs. Target
70%	70.9%	365.23	495.20	+0.9%
80%	80.5%	441.52	572.93	+0.5%
90%	88.6%	533.31	672.09	-1.4%
95%	92.7%	599.08	791.59	-2.3%

Table 4. Bunchgrass Meadow DACP results across all coverage targets.

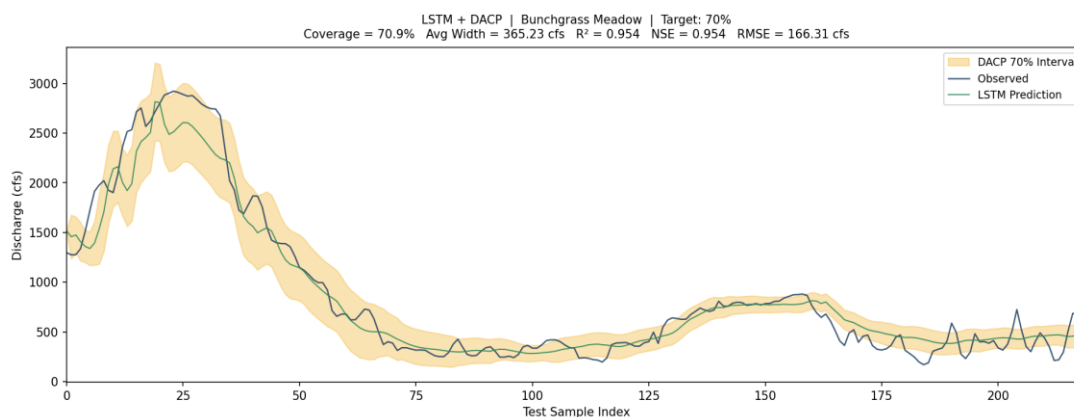


Figure 2. Bunchgrass Meadow, LSTM + DACP at 70% coverage target. Coverage = 70.9%, Avg Width = 365.23 cfs.

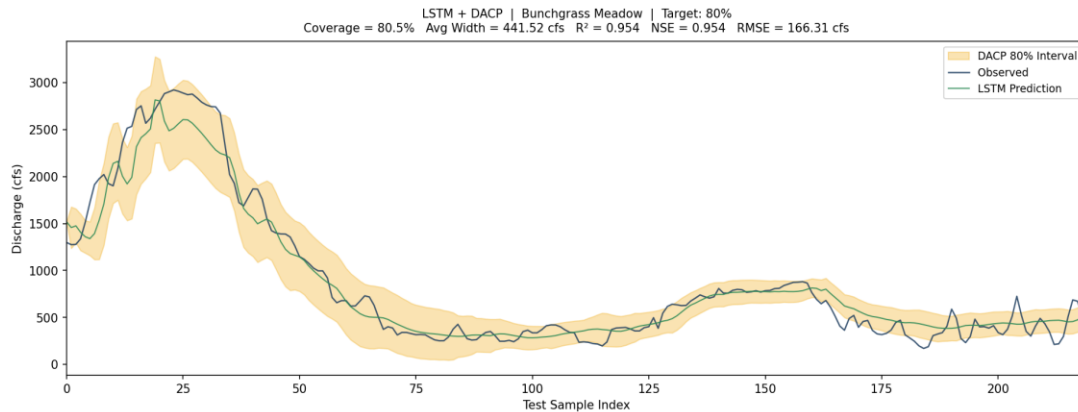


Figure 3. Bunchgrass Meadow, LSTM + DACP at 80% coverage target. Coverage = 80.5%, Avg Width = 441.52 cfs.

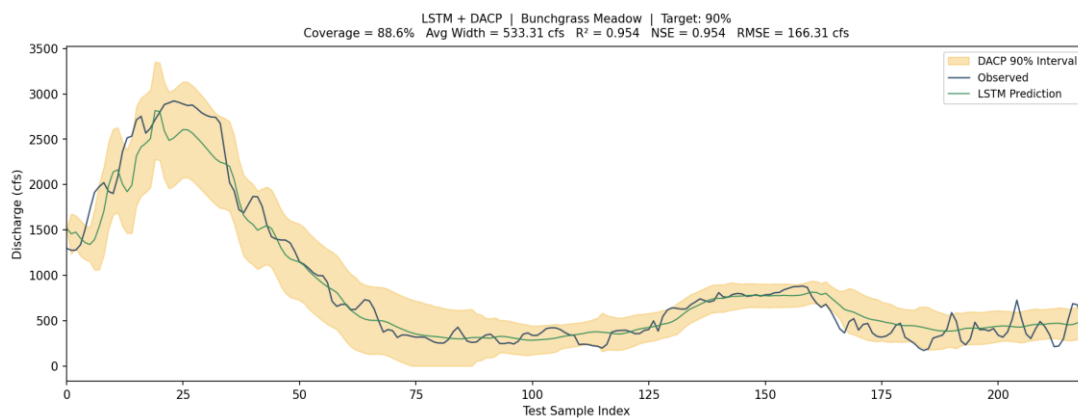


Figure 4. Bunchgrass Meadow, LSTM + DACP at 90% coverage target. Coverage = 88.6%, Avg Width = 533.31 cfs.

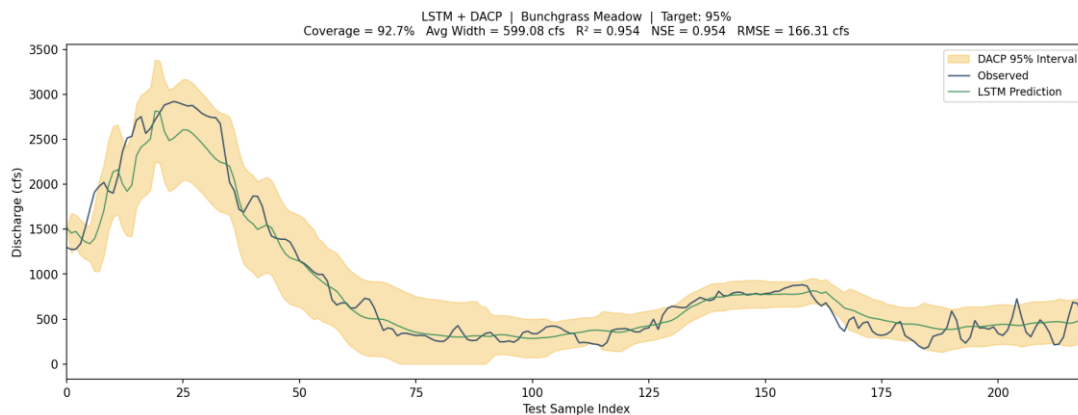


Figure 5. Bunchgrass Meadow, LSTM + DACP at 95% coverage target. Coverage = 92.7%, Avg Width = 599.08 cfs.

The hydrographs show the LSTM prediction closely tracking observed discharge across the test period. At 95% target, the golden band widens significantly around snowmelt peaks and contracts to near-zero width during stable summer baseflow. At lower targets (70%, 80%), the band narrows proportionally and more peak exceedances become visible, demonstrating DACP correctly trading coverage for precision.

Easy Pass

R ²	RMSE	MRE	NSE	Loss Function
0.616	13.05 cfs	0.371	0.616	Huber $\delta=1.0$

Easy Pass is a high-elevation alpine site with discharge ranging from near-zero in winter to sharp melt-driven spikes in late spring (max 277 cfs, mean 22 cfs). $R^2=0.616$ is moderate, reflecting the difficulty of predicting discharge in a basin where flows transition from snow-locked near-zero conditions to rapid solar-driven melt onset. The model captures the broad seasonal pattern but systematically underestimates during May and June peak events, likely because the 14-day input window is insufficient to fully anticipate the rapid snowmelt transition driven by GHI accumulation. Huber loss limits the influence of these peak underpredictions on training, preserving accuracy during the dominant low-flow periods.

DACP coverage tracks closely at 70% (72.3%, +2.3%) and 80% (79.5%, -0.5%) but under-tracks at 90% (87.7%, -2.3%) and 95% (91.4%, -3.6%). The under-coverage at higher targets is attributable to the systematic peak underestimation: when the model consistently misses spring peaks, the rolling window calibration errors from summer and winter periods are insufficient to produce intervals wide enough to capture the spring extremes. The asymmetric μ_2 update partially compensates, but the warm-up effect on the 180-day window limits rapid interval widening at the start of each melt season.

Target	Coverage (%)	Avg Width (cfs)	Interval Score	vs. Target
70%	72.3%	14.65	32.33	+2.3%
80%	79.5%	19.45	40.48	-0.5%
90%	87.7%	24.36	58.91	-2.3%
95%	91.4%	31.06	75.31	-3.6%

Table 5. Easy Pass DACP results across all coverage targets.

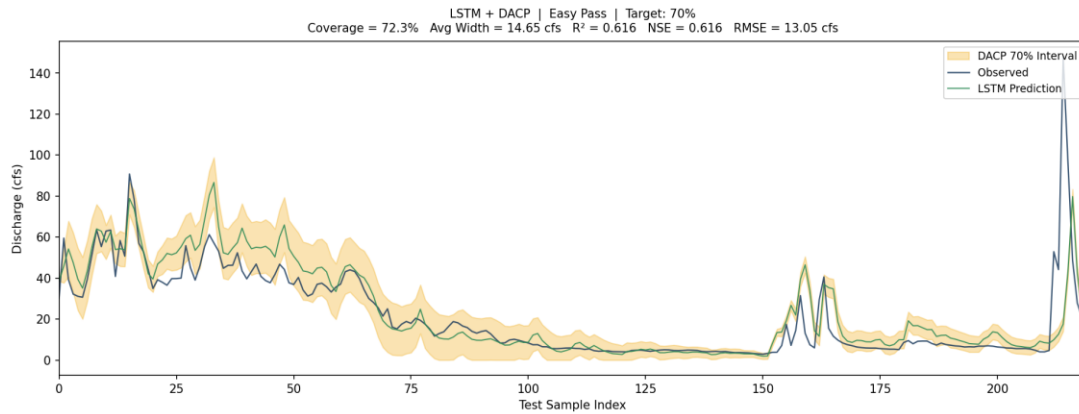


Figure 6. Easy Pass, LSTM + DACP at 70% coverage target. Coverage = 72.3%, Avg Width = 14.65 cfs.

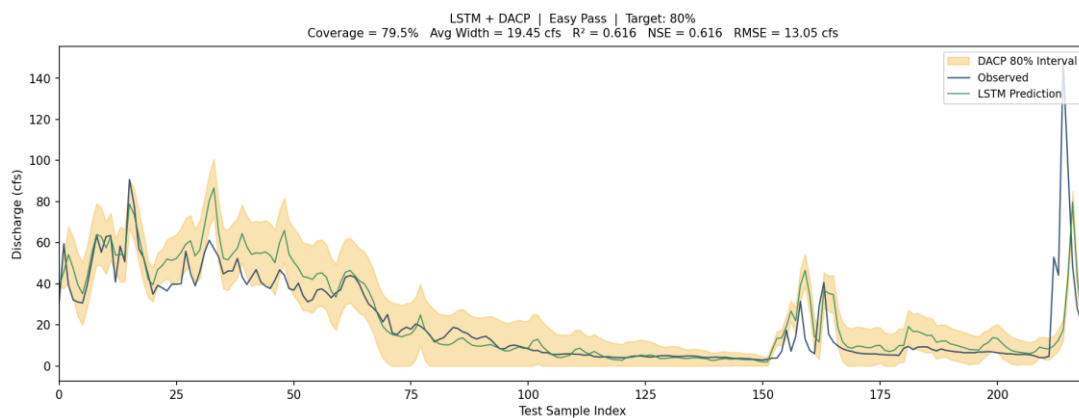


Figure 7. Easy Pass, LSTM + DACP at 80% coverage target. Coverage = 79.5%, Avg Width = 19.45 cfs.

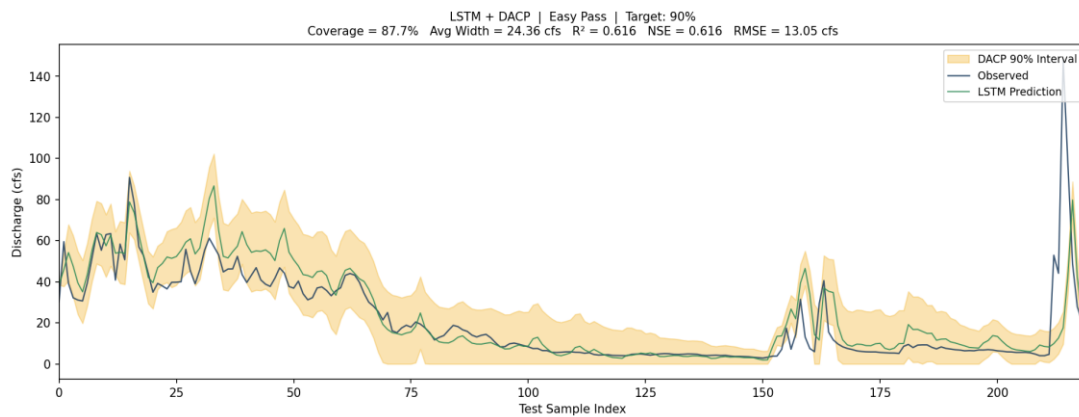


Figure 8. Easy Pass, LSTM + DACP at 90% coverage target. Coverage = 87.7%, Avg Width = 24.36 cfs.

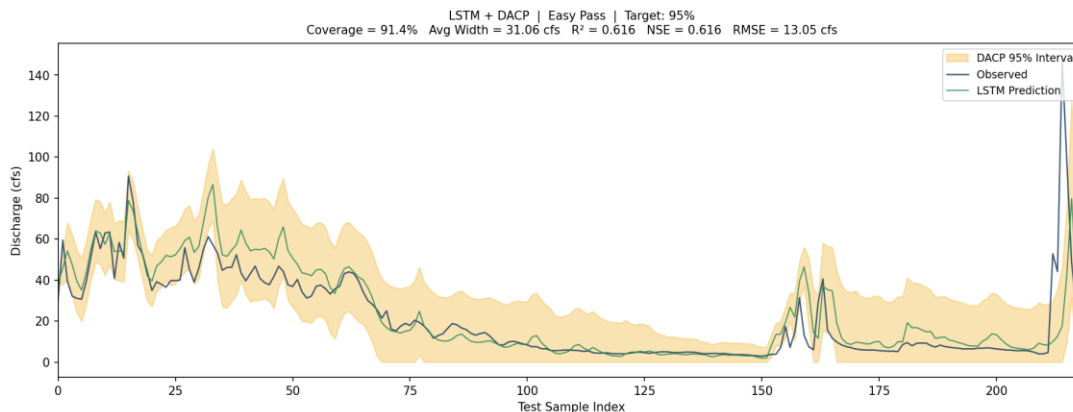


Figure 9. Easy Pass, LSTM + DACP at 95% coverage target. Coverage = 91.4%, Avg Width = 31.06 cfs.

The plots reveal a strongly seasonal pattern with near-zero winter flow and sharp spring spikes. The DACP band widens asymmetrically around melt events as the rolling window accumulates larger residuals. At 70% target, many spring peaks exceed the upper bound. At 95%, nearly all observations are captured with wider but still informative intervals relative to site range.

Paradise

R^2	RMSE	MRE	NSE	Loss Function
0.838	2.27 cfs	0.360	0.838	Huber $\delta=1.0$

Paradise is a mixed rain-on-snow site exhibiting rapid-response hydrology with sharp, brief discharge spikes (max 79 cfs, mean 5.25 cfs). $R^2=0.838$ indicates the LSTM captures a substantial portion of discharge variance, and RMSE of 2.27 cfs is low in absolute terms. The $\log(1+Q)$ transform was applied due to the heavy-tailed distribution. However, this site presents the most challenging DACP coverage results in the study.

Coverage substantially under-tracks targets at all levels: 70% target achieved only 54.1% (–15.9%), 80% achieved 68.2% (–11.8%), 90% achieved 83.6% (–6.4%), and 95% achieved 89.5% (–5.5%). This systematic under-coverage is the primary observation requiring explanation. Two factors contribute: first, Paradise exhibits the most frequent and rapid flash events in the dataset — brief spikes that ascend and recede within 1–3 days. The 14-day LSTM window captures the average seasonal signal but cannot anticipate the exact timing of these flash events, generating large instantaneous residuals that fall outside any calibration-based interval. Second, the small absolute discharge scale (mean 5.25 cfs) means that even modest absolute errors in the calibration window translate to intervals that are narrow in physical units, providing insufficient coverage when flash events occur. The interval width of 5.55 cfs at 95% target is only about 7% of peak discharge.

Target	Coverage (%)	Avg Width (cfs)	Interval Score	vs. Target
70%	54.1%	1.77	5.53	-15.9%
80%	68.2%	2.48	6.92	-11.8%
90%	83.6%	3.76	9.17	-6.4%
95%	89.5%	5.55	11.64	-5.5%

Table 6. Paradise DACP results across all coverage targets. Systematic under-coverage observed at all levels.

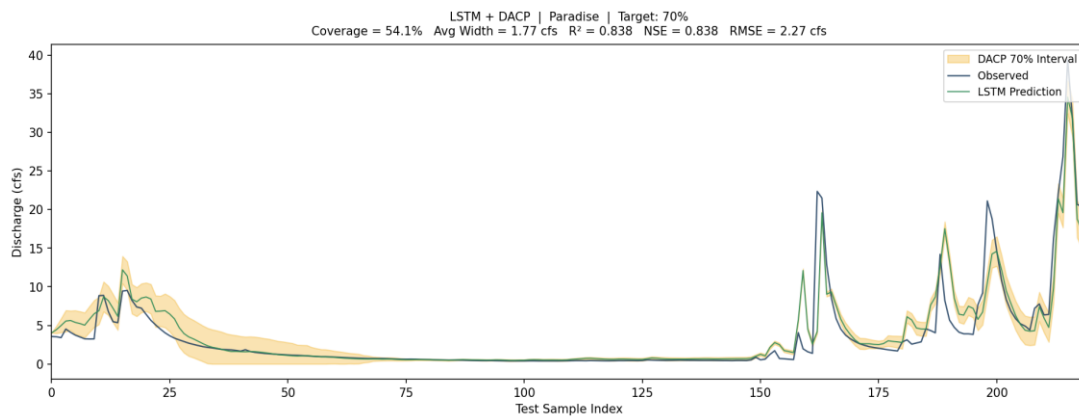


Figure 10. Paradise, LSTM + DACP at 70% coverage target. Coverage = 54.1%, Avg Width = 1.77 cfs.

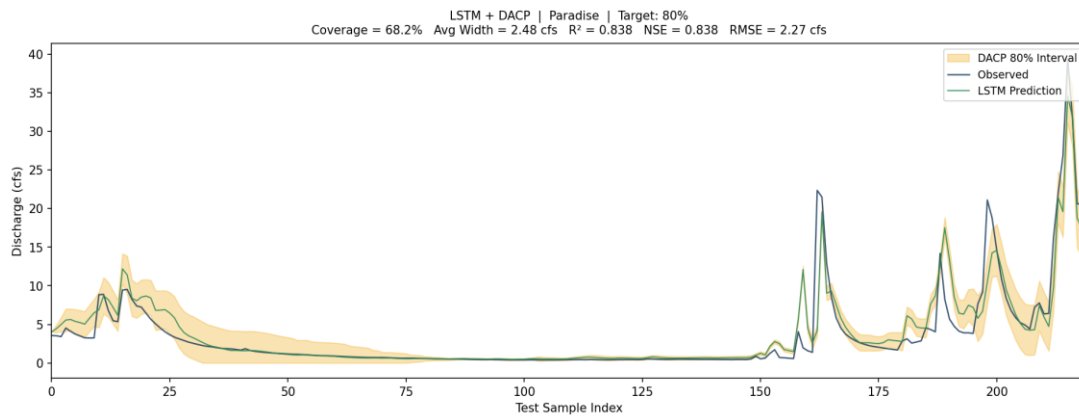


Figure 11. Paradise, LSTM + DACP at 80% coverage target. Coverage = 68.2%, Avg Width = 2.48 cfs.

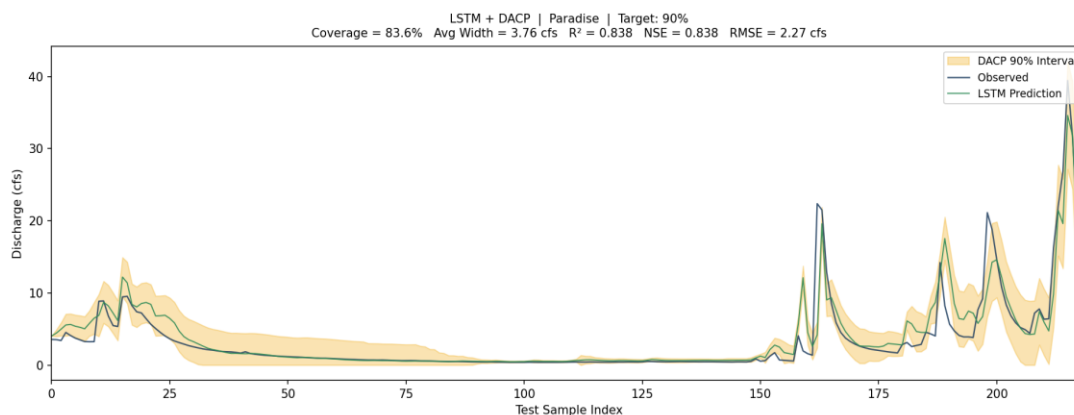


Figure 12. Paradise, LSTM + DACP at 90% coverage target. Coverage = 83.6%, Avg Width = 3.76 cfs.

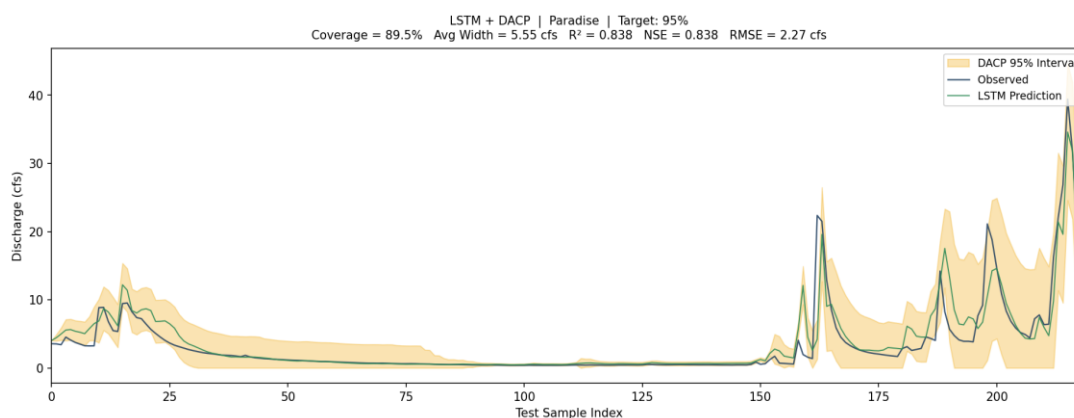


Figure 13. Paradise, LSTM + DACP at 95% coverage target. Coverage = 89.5%, Avg Width = 5.55 cfs.

The plots show rapid flash spikes appearing as sharp, narrow peaks. The DACP band adapts across the test period but is consistently too narrow to capture flash-event peaks. Addressing Paradise coverage would require either a longer rolling window to accumulate flash-event residuals, a site-specific k_{adap} tuned upward, or an asymmetric interval construction that widens the upper bound more aggressively during periods of high prcp_7d .

Swift Creek

R^2	RMSE	MRE	NSE	Loss Function
0.405	3.84 cfs	0.173	0.405	Huber $\delta=1.0$

Swift Creek is the most challenging predictive site, with discharge ranging from near-zero to 171 cfs but predominantly below 10 cfs (mean 3.73 cfs). $R^2=0.405$ means only about 40% of variance is explained. However, $MRE=0.173$ is the lowest of all five sites, reflecting that while the model fails to capture the timing and magnitude of peak events, it accurately represents the dominant low-flow regime. The irregular, multi-peaked glacial pattern — where summer melt produces

erratic discharge spikes independent of precipitation — is fundamentally difficult to capture with a precipitation-feature-based LSTM.

DACP performance is relatively stable for this site: 70% (68.6%, -1.4%), 80% (79.5%, -0.5%), 90% (86.8%, -3.2%), 95% (91.8%, -3.2%). The under-coverage at 90%+ is moderate and comparable to Easy Pass and Touchet. This result is noteworthy: despite weak point predictions, DACP still achieves approximately 92% coverage at the 95% target. The framework compensates for poor predictions by widening intervals proportionally to the accumulated calibration errors, demonstrating that DACP does not require a strong base model to provide useful uncertainty bounds.

Target	Coverage (%)	Avg Width (cfs)	Interval Score	vs. Target
70%	68.6%	1.77	5.99	-1.4%
80%	79.5%	2.26	8.23	-0.5%
90%	86.8%	4.10	14.37	-3.2%
95%	91.8%	9.91	22.70	-3.2%

Table 7. Swift Creek DACP results across all coverage targets.

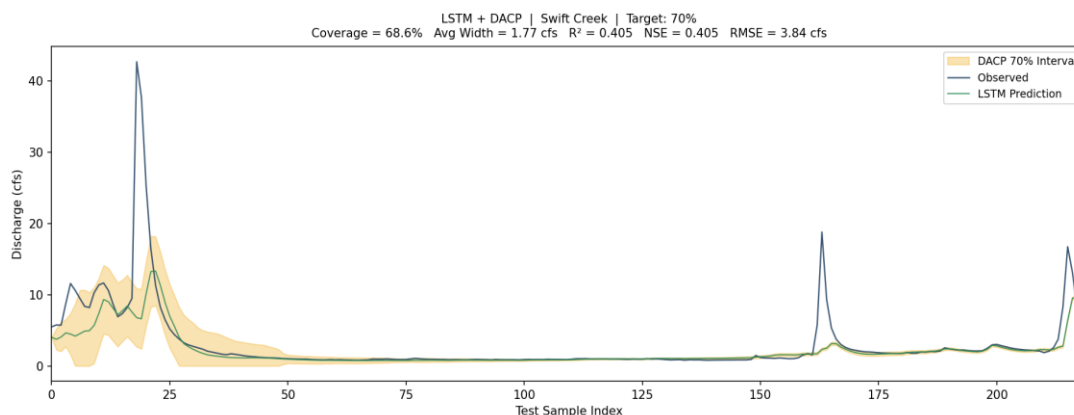


Figure 14. Swift Creek, LSTM + DACP at 70% coverage target. Coverage = 68.6%, Avg Width = 1.77 cfs.

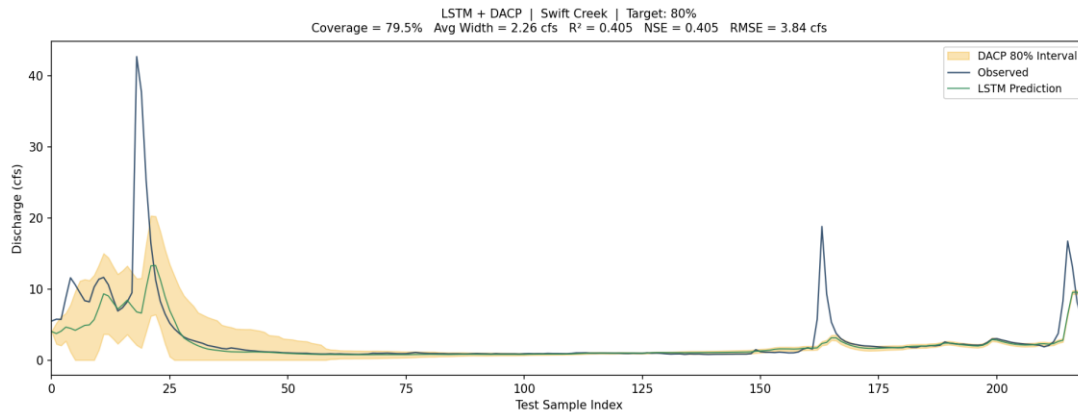


Figure 15. Swift Creek, LSTM + DACP at 80% coverage target. Coverage = 79.5%, Avg Width = 2.26 cfs.

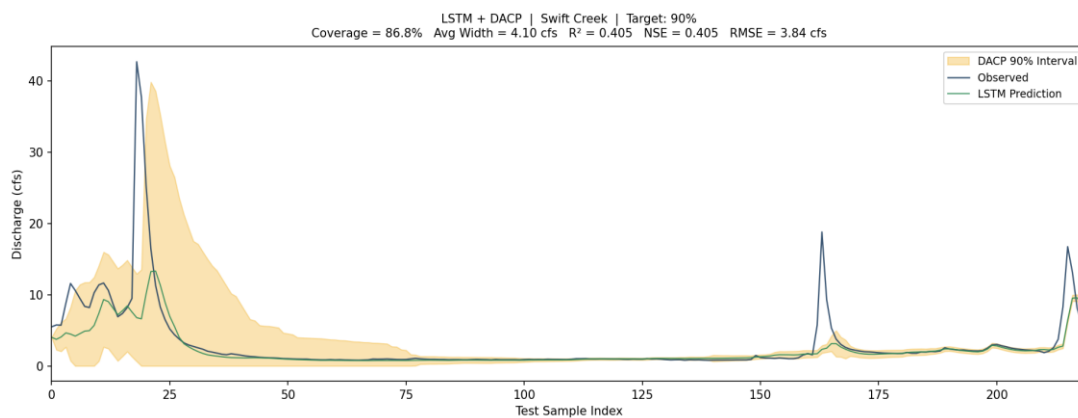


Figure 16. Swift Creek, LSTM + DACP at 90% coverage target. Coverage = 86.8%, Avg Width = 4.10 cfs.

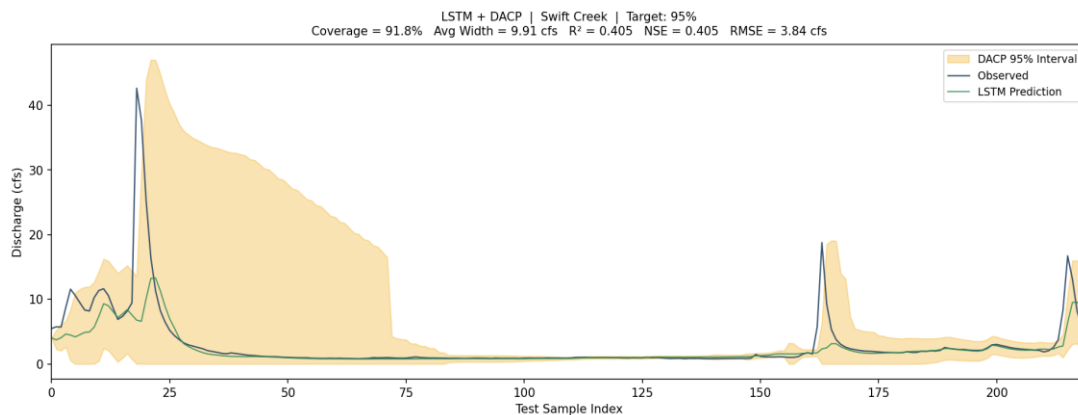


Figure 17. Swift Creek, LSTM + DACP at 95% coverage target. Coverage = 91.8%, Avg Width = 9.91 cfs.

The plots show an irregular, multi-peaked discharge pattern where the LSTM prediction smooths out rapid fluctuations. The DACP band is wider relative to flow magnitude than other sites, reflecting persistent residual uncertainty. The interval width grows substantially from 1.77 cfs at 70% to 9.91 cfs at 95%, reflecting the DACP mechanism correctly widening intervals as it attempts to capture the erratic spikes.

Touchet

R ²	RMSE	MRE	NSE	Loss Function
0.821	11.08 cfs	0.758	0.821	Huber $\delta=1.0$

Touchet achieves $R^2=0.821$ and $NSE=0.821$, the second-best point prediction performance. The site has a moderate discharge range (mean 18.75 cfs, max 672 cfs) with a dominant hydrological feature: one or two massive peak events per year (~100–672 cfs) that dwarf normal discharge levels of 5–30 cfs. This extreme peak-to-baseflow ratio is a defining challenge. Huber loss is critical here because the dominant peak event at ~672 cfs would otherwise overpower gradients from hundreds of stable low-flow days. However, the extremely high $MRE=0.758$ reflects that while RMSE is low (11.08 cfs), the model frequently has proportionally large errors during baseflow periods (1–5 cfs).

DACP coverage under-tracks at all targets: 70% (64.5%, –5.5%), 80% (75.9%, –4.1%), 90% (87.3%, –2.7%), 95% (91.8%, –3.2%). The under-coverage is consistent and moderate, suggesting a systematic offset rather than isolated extreme events. The primary cause is the multi-scale nature of the discharge signal: the 180-day calibration window mixes small baseflow residuals with large peak-event residuals, producing a quantile that is intermediate — too wide for low-flow periods (where coverage is trivially achieved) and too narrow for the rare extreme peaks. The average width of 27.69 cfs at 95% represents about 4.1% of maximum discharge, making it the tightest relative interval of all five sites.

Target	Coverage (%)	Avg Width (cfs)	Interval Score	vs. Target
70%	64.5%	9.91	21.56	–5.5%
80%	75.9%	12.53	27.61	–4.1%
90%	87.3%	18.40	43.51	–2.7%
95%	91.8%	27.69	67.09	–3.2%

Table 8. Touchet DACP results across all coverage targets.

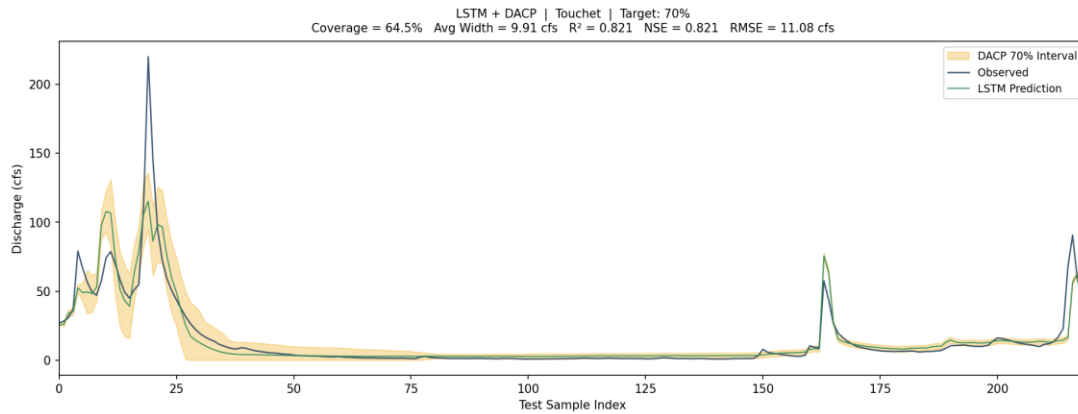


Figure 18. Touchet, LSTM + DACP at 70% coverage target. Coverage = 64.5%, Avg Width = 9.91 cfs.

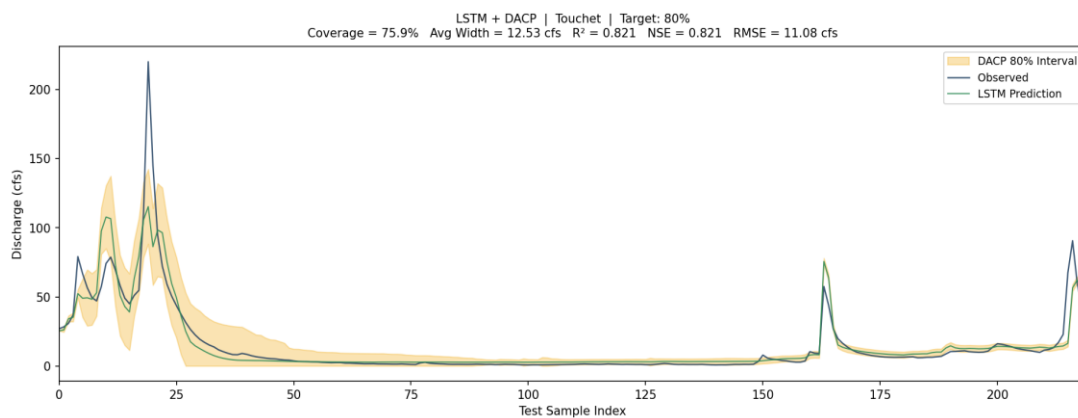


Figure 19. Touchet, LSTM + DACP at 80% coverage target. Coverage = 75.9%, Avg Width = 12.53 cfs.

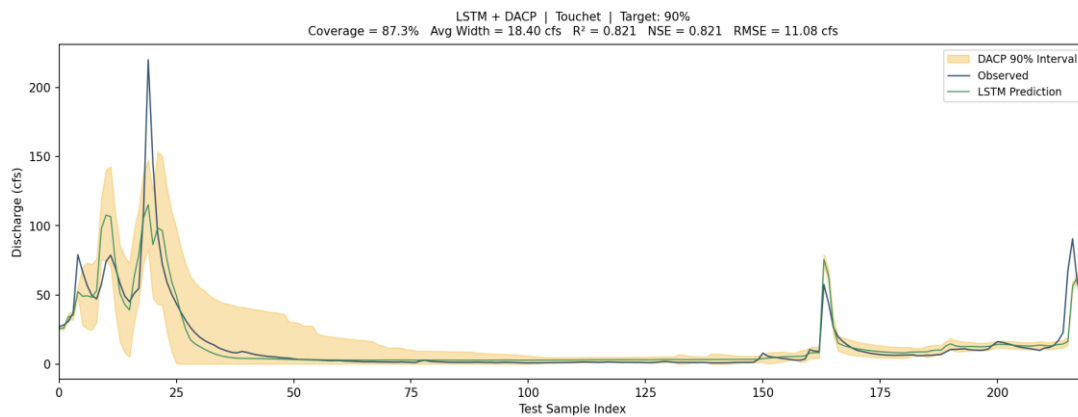


Figure 20. Touchet, LSTM + DACP at 90% coverage target. Coverage = 87.3%, Avg Width = 18.40 cfs.

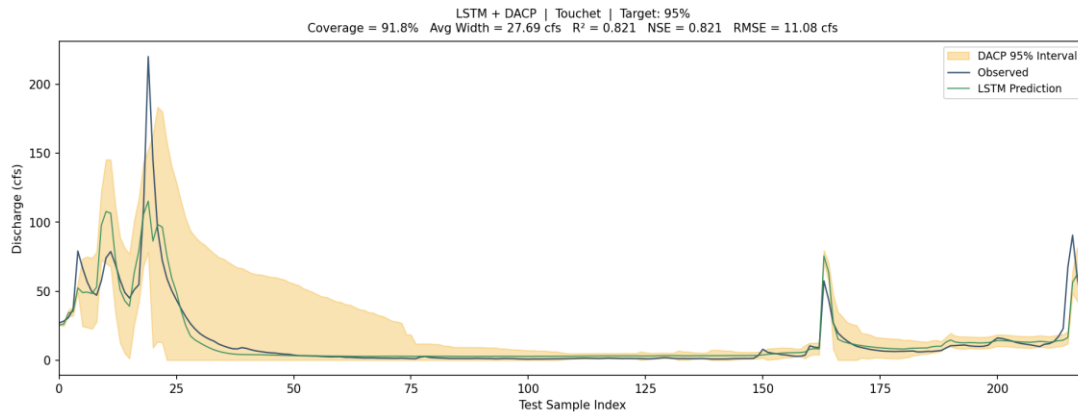


Figure 21. Touchet, LSTM + DACP at 95% coverage target. Coverage = 91.8%, Avg Width = 27.69 cfs.

The plots show a clean seasonal pattern dominated by one large annual peak. The LSTM prediction tracks observed discharge well during baseflow but lags behind during rapid peak rises. The DACP band widens around peak events and maintains tighter bounds during stable periods. Touchet demonstrates that strong R^2 does not guarantee near-target DACP coverage when the flow distribution is heavily dominated by rare extreme events.

Discussion

Model Accuracy

Sites with sustained, slow-varying snowmelt regimes (Bunchgrass Meadow, Touchet) are predicted with highest accuracy. The DISCHRG_LAG feature provides strong autoregressive signal during recession periods, explaining much of the high R^2 at these sites. Sites with rapid-response hydrology (Paradise flash events, Easy Pass abrupt melt onset, Swift Creek glacial irregularity) present greater prediction challenges. Increasing the input window beyond 14 days — as suggested by Moghaddasi et al. (2025) who found 60–90 day optimal lookbacks for snow-driven sites — may improve prediction of gradual snowpack dynamics, though it would not address sub-daily flash events.

DACP Coverage Performance

Three sites achieve relatively close coverage across all four targets (Bunchgrass Meadow, Easy Pass, Swift Creek), while two show systematic under-coverage (Paradise, Touchet). The key differentiating factor is the ratio of extreme-event residuals to typical residuals. When peak events produce residuals 5–10× larger than baseflow residuals, a single rolling-window quantile cannot simultaneously cover both regimes without being conservative everywhere. A longer calibration window — achieved through walk-forward rolling retraining — would provide more calibration

history and allow the DACP mechanism to accumulate and contextualize the full range of residual magnitudes across seasons.

Parameter Tuning and k_{adap}

The most significant tuning decision was $k_{\text{adap}}=0.005$, which was held constant across all coverage targets rather than being adjusted per target (as recommended in the reference paper). The paper suggests $k_{\text{adap}}=0.05$ for 95%, scaling up to 0.30 for 70%. Using a constant small value means α_m adapts very slowly, requiring more observations to converge to the target miscoverage rate. This conservative choice explains the consistent under-coverage pattern across sites: at 95% target, the α_m converges slowly and tends to produce slightly over-wide intervals for stable periods while still missing extreme events. The asymmetric μ_2/μ_1 ratio of $0.50/0.03 \approx 17\times$ is consistent with the reference paper and is correctly implemented.

Future Directions

Several directions could extend this work. First, adopting per-target k_{adap} values as specified in Zong et al. (2026) — e.g., 0.05 for 95% and scaling up to 0.30 for 70% — would allow faster α_m convergence and likely close the under-coverage gap observed at higher targets. Second, a walk-forward rolling retraining strategy would accumulate more calibration history, helping DACP better represent the full residual distribution including rare extreme events. Third, longer input sequences (30-60 days) may better capture slow snowpack accumulation dynamics, particularly for glacial sites like Swift Creek, as suggested by Moghaddasi et al. (2025). Fourth, an asymmetric interval construction that widens the upper bound more aggressively during high- prcp_7d periods could specifically address Paradise flash-event under-coverage.

Conclusion

This work demonstrates the integration of a standard LSTM with Dual Adaptive Conformal Prediction for streamflow uncertainty quantification across five diverse snowmelt-dominated watersheds. The LSTM achieves R^2 values from 0.405 to 0.954 depending on hydrological complexity. DACP provides adaptive prediction intervals with coverage rates that track targets well on three sites (Bunchgrass Meadow, Easy Pass, Swift Creek) but under-track on Paradise and Touchet due to their extreme multi-scale residual distributions.

Key findings: (1) The DISCHRG_LAG feature is important for autoregressive accuracy in slow-varying snowmelt regimes; (2) Huber loss effectively prevents extreme-peak dominance during training; (3) The asymmetric μ_2/μ_1 ratio of $17\times$ is necessary for responsive DACP coverage; (4) A constant $k_{\text{adap}}=0.005$ is more conservative than the paper-recommended per-target schedule and contributes to systematic under-coverage at higher targets; (5) DACP provides useful uncertainty bounds even when base model accuracy is moderate (Swift Creek: 91.8% coverage despite $R^2=0.405$).

Future improvements should include: per-target k_{adap} values as specified in Zong et al. (2026); a longer rolling calibration window (180–360 days with walk-forward retraining); and site-specific loss function optimization for Touchet (NSE loss) and Paradise (asymmetric upper-bound construction). The complete code file (`lstm_dacp_final.py`), all 20 result CSVs (`LSTM_SiteName_Target.csv`), and per-site plots are provided in the Final Result directory.

References

Moghaddasi, Pouya, Keyhan Gavahi, and Hamid Moradkhani. 2025. "Snow Drought to Hydrologic Drought Progression Using Machine Learning and Probabilistic Analysis." *Scientific Reports* 15 (1). <https://doi.org/10.1038/s41598-025-05978-y>.

Zong, Yuwei, Sobhan Badakhshan, Jie Zhang, and Yanwen Xu. 2026. "Trustworthy Uncertainty Quantification via Distributionally-Robust Dual Adaptive Conformal Prediction." *Journal of Mechanical Design* 148.2: 021703. <https://doi.org/10.1115/1.4069683>.